4.1 A 640×480 Dynamic Vision Sensor with a 9μm Pixel and 300Meps Address-Event Representation

Bongki Son[1], Yunjae Suh[1], Sungho Kim[1], Heejae Jung[1], Jun-Seok Kim[1], Changwoo Shin[1], Keunju Park[1], Kyoobin Lee[1], Jinman Park[1], Jooyeon Woo[1], Yohan Roh[1], Hyunku Lee[1], Yibing Wang[2], Ilia Ovsiannikov[2], Hyunsurk Ryu[1]

[1]Samsung Advanced Institute of Technology, Suwon, Korea
[2]Samsung Electronics, Pasadena, CA

We report a VGA dynamic vision sensor (DVS) with a 9μm pixel, developed through a digital as well as an analog implementation. DVS systems in the literature try to increase spatial resolution up to QVGA [1-2] and data rates up to 50 million events per second (Meps) (self-acknowledged) [3], but they are still inadequate for high-performance applications such as gesture recognition, drones, automotive, etc. Moreover, the smallest reported pixel of 18.5μm is too large for economical mass production [3]. This paper reports a 640×480 VGA-resolution DVS system with a 9μm pixel pitch supporting a data rate of 300Meps for sufficient event transfer in spite of higher resolution. Maintaining acceptable pixel performance, the pixel circuitry is carefully designed and optimized using a BSI CIS process. To acquire data (i.e., pixel events) at high speed even with high resolution (e.g., VGA), a fully synthesized word-serial group address-event representation (G-AER) is implemented, which handles massive events in parallel by binding neighboring 8 pixels into a group. In addition, a 10b programmable bias generator dedicated to a DVS system provides easy controllability of pixel biases and event thresholds.

Figure. 4.1.1 illustrates a DVS pixel circuit, composed of a gain-boosted log amplifier (LOGA), a source follower, a capacitive-feedback amplifier (CFA) followed by comparators, and 4-phase handshaking logic [3]. Previously, such circuit complexity had impeded the development of pixels smaller than $100\mu m^2$, while keeping the contrast sensitivity less than 10% due to a design tradeoff. This constraint is resolved by distributing pixel gain between LOGA and CFA and implementing the DVS in a BSI CIS process. The stacked transistor, M_{LOG1}, enhances the gain of the LOGA by (1+n), where n is the subthreshold slope factor, and thus leads the improvement in the contrast sensitivity in spite of decreasing the capacitive ratio, C_1/C_2, of CFA due to reduction in pixel pitch. Also, M_{N2} is cascoded to avoid BW degradation introduced by the additional pole at the drain node of M_{N2}. Furthermore, a conventional DVS pixel structure is vulnerable to periodic false events caused by the substrate leakage current from the power supply to the floating node, V_{IN}, via a reset transistor, M_{RESET}. Periodic false events, unnecessarily consuming AER bandwidth, are addressed by fetching the leakage path by tying the body of M_{RESET} to the output of CFA. The tied body could limit the output voltage swing when the drain-body junction of M_{RESET} is forward-biased, the event threshold voltages are set below 0.25V not to be affected by the limited swing. All current biases are supplied by a 10b programmable bias generator, also shown in Fig. 4.1.1. Pre-defined current sources are used for 3b coarse tuning, and 7b fine tuning is provided to achieve a wide dynamic range by the voltage drop across the programmable resistor bank in the feedback loop of amp. Since all bias transistors operate in the subthreshold region, I_{BIAS} exponentially increases along with an increase in 7b fine tuning code while V_{BIAS} linearly increases. Thus, a wide dynamic range is easily achieved without redundant bits.

Figure 4.1.2 shows the architecture of the DVS system with a 640×480 pixel array, a column AER, a row sampling circuit (RSC), a 10b programmable bias generator, and a fully synthesized G-AER. When events occur in pixels, the column AER sends a column request (ReqCol) and its corresponding column address (AddrX) to G-AER. G-AER sends an acknowledgement (AckCol) based on a handshaking logic and a whole 960-pixel data (ReqY_ON, ReqY_OFF) in the corresponding column are sent to RSC. Then, ReqY_ON and ReqY_OFF are sampled with AckCol and the sampled results (ON, OFF) are sent to G-AER. With these ON and OFF data, G-AER generates a 10b column address (AddrX_BUF), 6b group address (AddrYGrp), each 8b ON and OFF events (ONGrp, OFFGrp), and 32b time information (TimeStamp). The architectural timing diagram is also presented in Fig. 4.1.2.

Compared with previous works [1-6] that process events pixel-by-pixel with handshaking, which results in slow data throughput, G-AER can increase the AER data rate considerably. To increase the data rate up to 300Meps, several neighboring pixels in the column are grouped and dealt with like a single pixel, while ON and OFF events of the pixels in the same group are processed in parallel. Figure 4.1.3 illustrates the structures and the conceptual timing diagram of G-AER. The 480b ON and OFF events from RSC are divided into 60 groups by handling every neighboring 8 pixels as a unit and merged by 60 independent 16-input ORs whose output (GrpFlagIn) indicates whether any events exist in each group or not. The following FastFinder takes 60b GrpFlagIn and searches the location of 1-valued digit from LSB to MSB without redundant CLK cycle. Then, with the output of FastFinder (GrpFlagOut) at every CLK cycle, the group address encoder and group event selector output its corresponding group address and ON/OFF event data. The operating example of FastFinder including the output of G-AER is given in Figure 4.1.3. G-AER is fully synthesized at a timing constraint of 50MHz, thus it is possible to increase the data processing capacity over 300Meps by synthesizing at a timing constraint of more than 50MHz with no modification in the design.

The 640×480 DVS is implemented in a 90nm 1P5M BSI CIS process and occupies $8.0\times5.8mm^2$. The measured performance results are summarized in Fig. 4.1.6. The chip consumes a total of 27mW at a data rate of 100keps and 50mW at 300Meps. The power consumption per pixel is 88nW and 160nW at 100keps and 300Meps, respectively. Several measurement results are shown in Fig. 4.1.4. The rotating image with about 400rpm is captured by conventional CIS and DVS prototype cameras at the collecting time (Δt) of from 16.7 to 0.25ms. The letters on the DVS image are clear while they are blurred on the CIS image. Two images were taken at the different contrast sensitivities of 9% and 25%. (the second line from the bottom). The environment for testing dynamic range is shown at the left image on the bottom. The dynamic range is verified by checking that DVS can see bright and dark area at the same time. In result, right image shows DVS achieves wide dynamic range of >80dB. Figure 4.1.5 shows the measured transient data rate of the image recorded by DVS. The fixed image was taken with turning the light on and off. The chip photograph and pixel layout are shown in Fig. 4.1.7.

References:

[1] C. Li, et al., "An RGBW Color VGA Rolling and Global Shutter Dynamic and Active-Pixel Vision Sensor," *IEEE ISCAS,* pp. 718-721, 2015.
[2] C. Posch, et al., "A QVGA 143dB Dynamic Range Asynchronous Address-Event PWM Dynamic Image Sensor with Lossless Pixel-Level Video Compression," *ISSCC*, pp. 400-401, Feb. 2010.
[3] C. Brandli, et al., "A 240 × 180 130 dB 3 μs Latency Global Shutter Spatiotemporal Vision Sensor," *IEEE JSSC*, vol. 49, no. 10, pp. 2333-2341, 2014.
[4] T. Serrano-Gotarredona, et al., "A 128 × 128 1.5% Contrast Sensitivity 0.9% FPN 3 μs Latency 4 mW Asynchronous Frame Free Dynamic Vision Sensor Using Transimpedance Preamplifiers," *IEEE JSSC*, vol. 48, no. 3, pp. 827-838, 2013.
[5] J. A. Lenero-Bardallo, et al., "A 3.6μs Latency Asynchronous Frame-free Event-driven Dynamic-vision-sensor," *IEEE JSSC*, vol. 46, no. 6, pp. 1443-1455, 2011.
[6] P. Lichtsteiner, et al., "A 128×128 120dB 30mW Asynchronous Vision Sensor that Responds to Relative Intensity Change," *ISSCC*, pp. 508-509, Feb. 2006.

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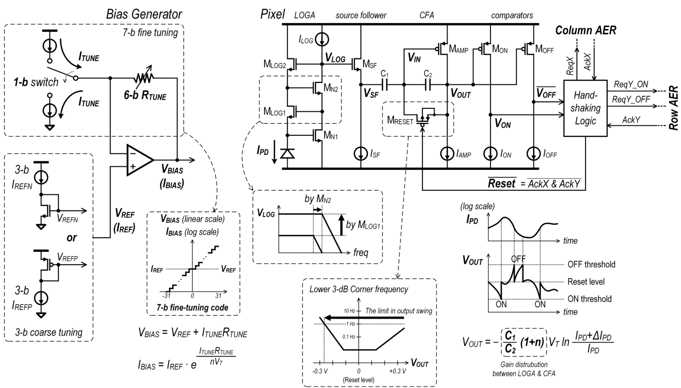

Figure 4.1.1: Pixel and bias generator.

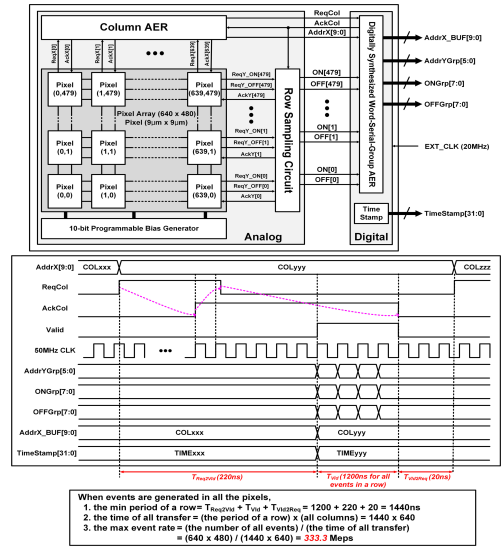

Figure 4.1.2: DVS architecture and timing diagram.

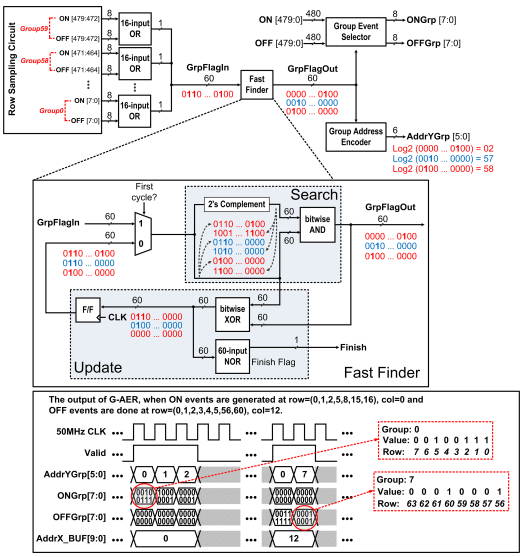

Figure 4.1.3: Diagrams of Group AER and FastFinder.

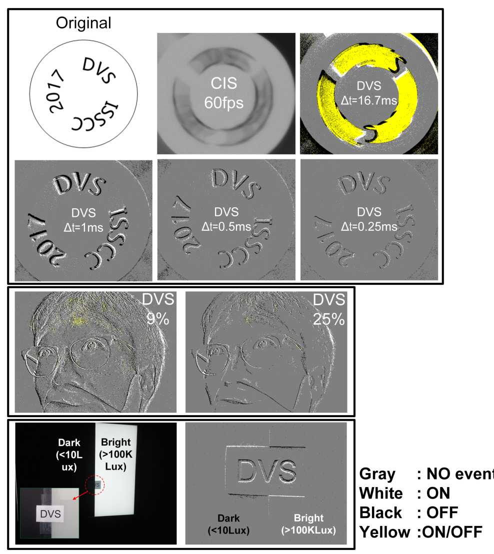

Figure 4.1.4: Speed, sensitivity, and WDR of DVS.

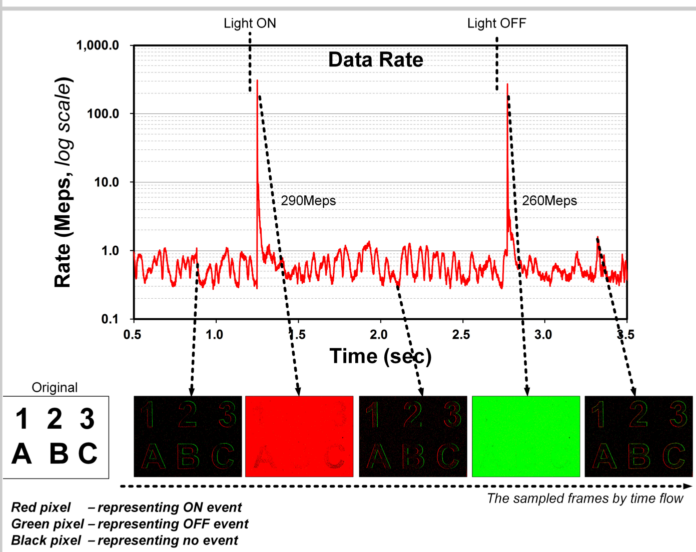

Figure 4.1.5: Transient data rate recorded by DVS.

		This work	JSSC'14 [3]	JSSC'13 [4]	JSSC'11 [5]	ISSCC'06 [6]
Technology		90nm 1P5M BSI	0.18μm 1P6M	0.35μm 2P4M	0.35μm 2P4M	0.35μm 2P4M
Resolution		640X480	240X180	128X128	128X128	128X128
Chip Area (mm^2)		8X5.8	5X5	4.9X4.9	5.5X5.6	6X6.3
Pixel Size (μm^2)		9X9	18.5X18.5	30X31	35X35	40X40
Power Supply		2.8V analog 1.2V digital	1.8V/3.3V (pixels)	3.3V	3.3V	3.3V
Power (mW)	100Keps	27[a]	14mW (high activity)	4	132	24
	300Meps	50[a]	5mW (low activity)	-	-	-
Power/Pixel (μW)[b]	100Keps	0.088	0.32 (high activity)	0.24	8.06	1.46
	300Meps	0.16	0.12 (low activity)	-	-	-
Power/Event Rate (nW)[c]	100Keps	270[d]	-	40	1320	240
	300Meps	0.17		-	-	-
Min. Contrast Sensitivity (%)		9	11	1.5	10	15
Max. Event Rate (Meps)		300	50 (self ack) 12 (CPLD)	20	20	1
Chip Interface		MIPI, USB Connectable I/F, I2C	Handshake	Handshake	Handshake	Handshake

a. Measured average power of 30 samples. PLL power (9mW) is included.
b. Total power normalized to resolution
c. Total power normalized to max. event rate
d. The total power of this work includes IO power and digital interface. Others do not.

Figure 4.1.6: Comparison with state of the art.

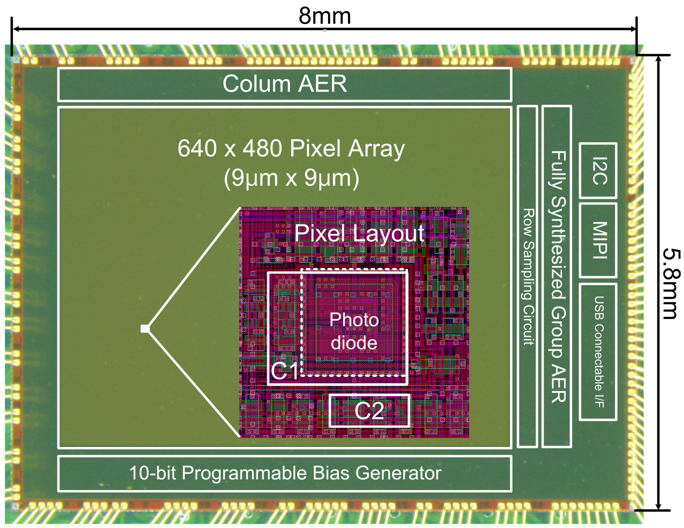

Figure 4.1.7: Micrograph of DVS chip in BSI process.

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